

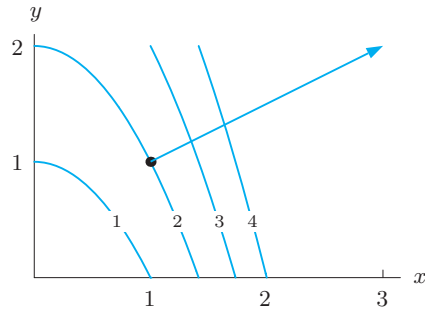


Figure 14.46: The gradient vector with  $x$  and  $y$  scales equal

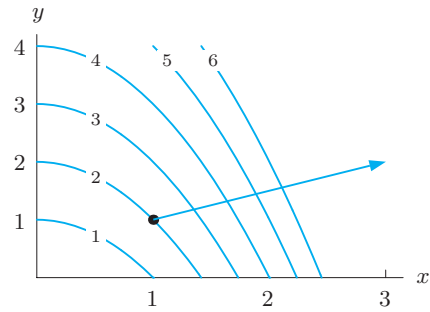


Figure 14.47: The gradient vector with  $x$  and  $y$  scales unequal

## Exercises and Problems for Section 14.5

### Exercises

In Exercises 1–12, find the gradient of the function.

- $f(x, y, z) = x^2$
- $f(x, y, z) = x^2 + y^3 - z^4$
- $f(x, y, z) = e^{x+y+z}$
- $f(x, y, z) = \cos(x+y) + \sin(y+z)$
- $f(x, y, z) = yz^2/(1+x^2)$
- $f(x, y, z) = 1/(x^2 + y^2 + z^2)$
- $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$
- $f(x, y, z) = xe^{yz} \sin z$
- $f(x, y, z) = xy + \sin(e^z)$
- $f(x_1, x_2, x_3) = x_1^2 x_2^3 x_3^4$
- $f(p, q, r) = e^p + \ln q + e^{r^2}$
- $f(x, y, z) = e^{z^2} + y \ln(x^2 + 5)$

In Exercises 13–18, find the gradient at the point.

- $f(x, y, z) = zy^2$ , at  $(1, 0, 1)$
- $f(x, y, z) = 2x + 3y + 4z$ , at  $(1, 1, 1)$
- $f(x, y, z) = x^2 + y^2 - z^4$ , at  $(3, 2, 1)$
- $f(x, y, z) = xyz$ , at  $(1, 2, 3)$
- $f(x, y, z) = \sin(xy) + \sin(yz)$ , at  $(1, \pi, -1)$
- $f(x, y, z) = x \ln(yz)$ , at  $(2, 1, e)$

In Exercises 19–24, find the directional derivative using  $f(x, y, z) = xy + z^2$ .

- At  $(1, 2, 3)$  in the direction of  $\vec{i} + \vec{j} + \vec{k}$ .
- At  $(1, 1, 1)$  in the direction of  $\vec{i} + 2\vec{j} + 3\vec{k}$ .
- As you leave the point  $(1, 1, 0)$  heading in the direction of the point  $(0, 1, 1)$ .
- As you arrive at  $(0, 1, 1)$  from the direction of  $(1, 1, 0)$ .
- At the point  $(2, 3, 4)$  in the direction of a vector making an angle of  $3\pi/4$  with  $\text{grad } f(2, 3, 4)$ .
- At the point  $(2, 3, 4)$  in the direction of the maximum rate of change of  $f$ .

In Exercises 25–30, check that the point  $(-1, 1, 2)$  lies on the given surface. Then, viewing the surface as a level surface for a function  $f(x, y, z)$ , find a vector normal to the surface and an equation for the tangent plane to the surface at  $(-1, 1, 2)$ .

- $x^2 - y^2 + z^2 = 4$
- $z = x^2 + y^2$
- $y^2 = z^2 - 3$
- $x^2 - xyz = 3$
- $\cos(x+y) = e^{xz+2}$
- $y = 4/(2x+3z)$
- For  $f(x, y, z) = 3x^2y^2 + 2yz$ , find the directional derivative at the point  $(-1, 0, 4)$  in the direction of
  - $\vec{i} - \vec{k}$
  - $-\vec{i} + 3\vec{j} + 3\vec{k}$
- If  $f(x, y, z) = x^2 + 3xy + 2z$ , find the directional derivative at the point  $(2, 0, -1)$  in the direction of  $2\vec{i} + \vec{j} - 2\vec{k}$ .
- (a) Let  $f(x, y, z) = x^2 + y^2 - xyz$ . Find  $\text{grad } f$ .  
(b) Find the equation for the tangent plane to the surface  $f(x, y, z) = 7$  at the point  $(2, 3, 1)$ .
- Find the equation of the tangent plane at the point  $(3, 2, 2)$  to  $z = \sqrt{17 - x^2 - y^2}$ .
- Find the equation of the tangent plane to  $z = 8/(xy)$  at the point  $(1, 2, 4)$ .
- Find an equation of the tangent plane and of a normal vector to the surface  $x = y^3z^7$  at the point  $(1, -1, -1)$ .

In Exercises 37–38, the gradient of  $f$  and a point  $P$  on the level surface  $f(x, y, z) = 0$  are given. Find an equation for the tangent plane to the surface at the point  $P$ .

- $\text{grad } f = yz\vec{i} + xz\vec{j} + xy\vec{k}$ ,  $P = (1, 2, 3)$
- $\text{grad } f = 2x\vec{i} + z^2\vec{j} + 2yz\vec{k}$ ,  $P = (10, -10, 30)$

In Exercises 39–43, find an equation of the tangent plane to the surface at the given point.

- $x^2 + y^2 + z^2 = 17$  at the point  $(2, 3, 2)$
- $x^2 + y^2 = 1$  at the point  $(1, 0, 0)$
- $z = 2x + y + 3$  at the point  $(0, 0, 3)$
- $3x^2 - 4xy + z^2 = 0$  at the point  $(a, a, a)$ , where  $a \neq 0$
- $z = 9/(x + 4y)$  at the point where  $x = 1$  and  $y = 2$

## Problems

44. Consider the surface  $g(x, y) = 4 - x^2$ . What is the relation between  $\text{grad } g(-1, -1)$  and a vector tangent to the path of steepest ascent at  $(-1, -1, 3)$ ? Illustrate your answer with a sketch.

45. Match the functions  $f(x, y, z)$  in (a)–(d) with the descriptions of their gradients in (I)–(IV). No reasons needed.

- (a)  $x^2 + y^2 + z^2$  (b)  $x^2 + y^2$   
 (c)  $\frac{1}{x^2 + y^2 + z^2}$  (d)  $\frac{1}{x^2 + y^2}$

- I Points radially outward from the  $z$ -axis.  
 II Points radially inward toward the  $z$ -axis.  
 III Points radially outward from the origin.  
 IV Points radially inward toward the origin.

46. Find the equation of the tangent plane at  $(2, 3, 1)$  to the surface  $x^2 + y^2 - xyz = 7$ . Do this in two ways:

- (a) Viewing the surface as the level set of a function of three variables,  $F(x, y, z)$ .  
 (b) Viewing the surface as the graph of a function of two variables  $z = f(x, y)$ .

47. Let  $f(x, y, z) = x^2 + y^2 + z^2$ . At the point  $(1, 2, 1)$ , find the rate of change of  $f$  in the direction perpendicular to the plane  $x + 2y + 3z = 8$  and moving away from the origin.

48. Let  $f(x, y) = \cos x \sin y$  and let  $S$  be the surface  $z = f(x, y)$ .

- (a) Find a normal vector to the surface  $S$  at the point  $(0, \pi/2, 1)$ .  
 (b) What is the equation of the tangent plane to the surface  $S$  at the point  $(0, \pi/2, 1)$ ?

49. Let  $f(x, y, z) = \sin(x^2 + y^2 + z^2)$ .

- (a) Describe in words the shape of the level surfaces of  $f$ .  
 (b) Find  $\text{grad } f$ .  
 (c) Consider the two vectors  $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$  and  $\text{grad } f$  at a point  $(x, y, z)$  where  $\sin(x^2 + y^2 + z^2) \neq 0$ . What is (are) the possible value(s) of the angle between these vectors?

50. Each diagram (I) – (IV) in Figure 14.48 represents the level curves of a function  $f(x, y)$ . For each function  $f$ , consider the point above  $P$  on the surface  $z = f(x, y)$  and choose from the lists which follow:

- (a) A vector which could be the normal to the surface at that point;  
 (b) An equation which could be the equation of the tangent plane to the surface at that point.

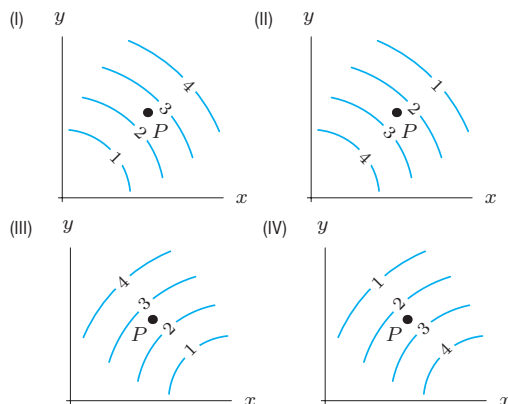


Figure 14.48

- | Vectors                               | Equations                                           |
|---------------------------------------|-----------------------------------------------------|
| (E) $2\vec{i} + 2\vec{j} - 2\vec{k}$  | (J) $x + y + z = 4$                                 |
| (F) $2\vec{i} + 2\vec{j} + 2\vec{k}$  | (K) $2x - 2y - 2z = 2$                              |
| (G) $2\vec{i} - 2\vec{j} + 2\vec{k}$  | (L) $-3x - 3y + 3z = 6$                             |
| (H) $-2\vec{i} + 2\vec{j} + 2\vec{k}$ | (M) $-\frac{x}{2} + \frac{y}{2} - \frac{z}{2} = -7$ |

51. The surface  $S$  is represented by the equation  $F = 0$  where  $F(x, y, z) = x^2 - (y/z^2)$ .

- (a) Find the unit vectors  $\vec{u}_1$  and  $\vec{u}_2$  pointing in the direction of maximum increase of  $F$  at the points  $(0, 0, 1)$  and  $(1, 1, 1)$  respectively.  
 (b) Find the tangent plane to  $S$  at the points  $(0, 0, 1)$  and  $(1, 1, 1)$ .  
 (c) Find all points on  $S$  where a normal vector is parallel to the  $xy$ -plane.

52. Consider the function  $f(x, y) = (e^x - x) \cos y$ . Suppose  $S$  is the surface  $z = f(x, y)$ .

- (a) Find a vector which is perpendicular to the level curve of  $f$  through the point  $(2, 3)$  in the direction in which  $f$  decreases most rapidly.  
 (b) Suppose  $\vec{v} = 5\vec{i} + 4\vec{j} + a\vec{k}$  is a vector in 3-space which is tangent to the surface  $S$  at the point  $P$  lying on the surface above  $(2, 3)$ . What is  $a$ ?

53. (a) Find the tangent plane to the surface  $x^2 + y^2 + 3z^2 = 4$  at the point  $(0.6, 0.8, 1)$ .

- (b) Is there a point on the surface  $x^2 + y^2 + 3z^2 = 4$  at which the tangent plane is parallel to the plane  $8x + 6y + 30z = 1$ ? If so, find it. If not, explain why not.

54. Your house lies on the surface  $z = f(x, y) = 2x^2 - y^2$  directly above the point  $(4, 3)$  in the  $xy$ -plane.

- (a) How high above the  $xy$ -plane do you live?  
 (b) What is the slope of your lawn as you look from your house directly toward the  $z$ -axis (that is, along the vector  $-4\vec{i} - 3\vec{j}$ )?

- (c) When you wash your car in the driveway, on this surface above the point  $(4, 3)$ , which way does the water run off? (Give your answer as a two-dimensional vector.)
- (d) What is the equation of the tangent plane to this surface at your house?
55. (a) Sketch the contours of  $z = y - \sin x$  for  $z = -1, 0, 1, 2$ .
- (b) A bug starts on the surface at the point  $(\pi/2, 1, 0)$  and walks on the surface  $z = y - \sin x$  in the direction parallel to the  $y$ -axis, in the direction of increasing  $y$ . Is the bug walking in a valley or on top of a ridge? Explain.
- (c) On the contour  $z = 0$  in your sketch for part (a), draw the gradients of  $z$  at  $x = 0$ ,  $x = \pi/2$ , and  $x = \pi$ .
56. At what point on the surface  $z = 1 + x^2 + y^2$  is its tangent plane parallel to the following planes?
- (a)  $z = 5$  (b)  $z = 5 + 6x - 10y$ .
57. The concentration of salt in a fluid at  $(x, y, z)$  is given by  $F(x, y, z) = x^2 + y^4 + x^2 z^2$  mg/cm<sup>3</sup>. You are at the point  $(-1, 1, 1)$ .
- (a) In which direction should you move if you want the concentration to increase the fastest?
- (b) You start to move in the direction you found in part (a) at a speed of 4 cm/sec. How fast is the concentration changing?
58. Let  $g_x(2, 1, 7) = 3$ ,  $g_y(2, 1, 7) = 10$ ,  $g_z(2, 1, 7) = -5$ . Find the equation of the tangent plane to  $g(x, y, z) = 0$  at the point  $(2, 1, 7)$ .
59. The vector  $\nabla f$  at point  $P$  and four unit vectors  $\vec{u}_1, \vec{u}_2, \vec{u}_3, \vec{u}_4$  are shown in Figure 14.49. Arrange the following quantities in ascending order
- $f_{\vec{u}_1}, f_{\vec{u}_2}, f_{\vec{u}_3}, f_{\vec{u}_4},$  the number 0.

The directional derivatives are all evaluated at the point  $P$  and the function  $f(x, y)$  is differentiable at  $P$ .

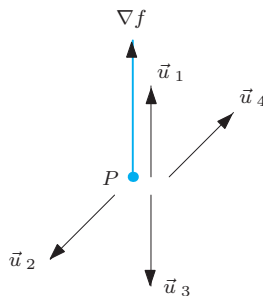


Figure 14.49

60. The temperature of a gas at the point  $(x, y, z)$  is given by  $G(x, y, z) = x^2 - 5xy + y^2 z$ .
- (a) What is the rate of change in the temperature at the point  $(1, 2, 3)$  in the direction  $\vec{v} = 2\vec{i} + \vec{j} - 4\vec{k}$ ?
- (b) What is the direction of maximum rate of change of temperature at the point  $(1, 2, 3)$ ?
- (c) What is the maximum rate of change at the point  $(1, 2, 3)$ ?
61. The temperature at the point  $(x, y, z)$  in 3-space is given, in degrees Celsius, by  $T(x, y, z) = e^{-(x^2 + y^2 + z^2)}$ .
- (a) Describe in words the shape of surfaces on which the temperature is constant.
- (b) Find  $\text{grad } T$ .
- (c) You travel from the point  $(1, 0, 0)$  to the point  $(2, 1, 0)$  at a speed of 3 units per second. Find the instantaneous rate of change of the temperature as you leave the point  $(1, 0, 0)$ . Give units.
62. A spaceship is plunging into the atmosphere of a planet. With coordinates in miles and the origin at the center of the planet, the pressure of the atmosphere at  $(x, y, z)$  is
- $$P = 5e^{-0.1\sqrt{x^2 + y^2 + z^2}} \text{ atmospheres.}$$
- The velocity, in miles/sec, of the spaceship at  $(0, 0, 1)$  is  $\vec{v} = \vec{i} - 2.5\vec{k}$ . At  $(0, 0, 1)$ , what is the rate of change with respect to time of the pressure on the spaceship?
63. Let  $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$  and  $\vec{a}$  be a constant vector. For each of the quantities in (a)–(c), choose the statement in (I)–(V) that describes it. No reasons needed.
- (a)  $\text{grad}(\vec{r} \cdot \vec{a})$  (b)  $\text{grad}(\vec{r} \cdot \vec{a})$  (c)  $\text{grad}(\vec{r} \times \vec{a})$
- I Scalar, independent of  $\vec{a}$ .  
 II Scalar, depends on  $\vec{a}$ .  
 III Vector, independent of  $\vec{a}$ .  
 IV Vector, depends on  $\vec{a}$ .  
 V Not defined.
64. The earth has mass  $M$  and is located at the origin in 3-space, while the moon has mass  $m$ . Newton's Law of Gravitation states that if the moon is located at the point  $(x, y, z)$  then the attractive force exerted by the earth on the moon is given by the vector

$$\vec{F} = -GMm \frac{\vec{r}}{\|\vec{r}\|^3},$$

where  $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ . Show that  $\vec{F} = \text{grad } \varphi$ , where  $\varphi$  is the function given by

$$\varphi(x, y, z) = \frac{GMm}{\|\vec{r}\|}.$$

65. Two surfaces are said to be *tangential* at a point  $P$  if they have the same tangent plane at  $P$ . Show that the surfaces  $z = \sqrt{2x^2 + 2y^2 - 25}$  and  $z = \frac{1}{5}(x^2 + y^2)$  are tangential at the point  $(4, 3, 5)$ .

66. Two surfaces are said to be *orthogonal* to each other at a point  $P$  if the normals to their tangent planes are perpendicular at  $P$ . Show that the surfaces  $z = \frac{1}{2}(x^2 + y^2 - 1)$  and  $z = \frac{1}{2}(1 - x^2 - y^2)$  are orthogonal at all points of intersection.

67. Let  $\vec{r}$  be the position vector of the point  $(x, y, z)$ . If  $\vec{\mu} = \mu_1\vec{i} + \mu_2\vec{j} + \mu_3\vec{k}$  is a constant vector, show that

$$\text{grad}(\vec{\mu} \cdot \vec{r}) = \vec{\mu}.$$

68. Let  $\vec{r}$  be the position vector of the point  $(x, y, z)$ . Show

that, if  $a$  is a constant,

$$\text{grad}(\|\vec{r}\|^a) = a\|\vec{r}\|^{a-2}\vec{r}, \quad \vec{r} \neq \vec{0}.$$

69. Let  $f$  and  $g$  be functions on 3-space. Suppose  $f$  is differentiable and that

$$\text{grad } f(x, y, z) = (x\vec{i} + y\vec{j} + z\vec{k})g(x, y, z).$$

Explain why  $f$  must be constant on any sphere centered at the origin.

### Strengthen Your Understanding

In Problems 70–71, explain what is wrong with the statement.

70. The gradient vector  $\text{grad } f(x, y)$  points in the direction perpendicular to the surface  $z = f(x, y)$ .

71. The tangent plane at the origin to a surface  $f(x, y, z) = 1$  that contains the point  $(0, 0, 0)$  has equation

$$f_x(0, 0, 0)x + f_y(0, 0, 0)y + f_z(0, 0, 0)z + 1 = 0.$$

In Problems 72–74, give an example of:

72. A surface  $z = f(x, y)$  such that the vector  $\vec{i} - 2\vec{j} - \vec{k}$  is normal to the tangent plane at the point where  $(x, y) = (0, 0)$ .

73. A function  $f(x, y, z)$  such that  $\text{grad } f = 2\vec{i} + 3\vec{j} + 4\vec{k}$ .

74. Two nonparallel unit vectors  $\vec{u}$  and  $\vec{v}$  such that  $f_{\vec{u}}(0, 0, 0) = f_{\vec{v}}(0, 0, 0) = 0$ , where  $f(x, y, z) = 2x - 3y$ .

Are the statements in Problems 75–78 true or false? Give reasons for your answer.

75. An equation for the tangent plane to the surface  $z = x^2 + y^3$  at  $(1, 1)$  is  $z = 2 + 2x(x - 1) + 3y^2(y - 1)$ .

76. There is a function  $f(x, y)$  which has a tangent plane with equation  $z = 0$  at a point  $(a, b)$ .

77. There is a function with  $\|\text{grad } f\| = 4$  and  $f_{\vec{k}} = 5$  at some point.

78. There is a function with  $\|\text{grad } f\| = 5$  and  $f_{\vec{k}} = -3$  at some point.

79. Let  $f(x, y, z)$  represent the temperature in  $^{\circ}\text{C}$  at the point  $(x, y, z)$  with  $x, y, z$  in meters. Let  $\vec{v}$  be your velocity in meters per second. Give units and an interpretation of each of the following quantities.

$$(a) \quad \|\text{grad } f\| \quad (b) \quad \text{grad } f \cdot \vec{v} \quad (c) \quad \|\text{grad } f\| \cdot \|\vec{v}\|$$

## 14.6 THE CHAIN RULE

### Composition of Functions of Many Variables and Rates of Change

The chain rule enables us to differentiate *composite functions*. If we have a function of two variables  $z = f(x, y)$  and we substitute  $x = g(t)$ ,  $y = h(t)$  into  $z = f(x, y)$ , then we have a composite function in which  $z$  is a function of  $t$ :

$$z = f(g(t), h(t)).$$

If, on the other hand, we substitute  $x = g(u, v)$ ,  $y = h(u, v)$ , then we have a different composite function in which  $z$  is a function of  $u$  and  $v$ :

$$z = f(g(u, v), h(u, v)).$$

The next example shows how to calculate the rate of change of a composite function.

**Example 1** Corn production,  $C$ , depends on annual rainfall,  $R$ , and average temperature,  $T$ , so  $C = f(R, T)$ . Global warming predicts that both rainfall and temperature depend on time. Suppose that according to a particular model of global warming, rainfall is decreasing at 0.2 cm per year and temperature is increasing at 0.1 $^{\circ}\text{C}$  per year. Use the fact that at current levels of production,  $f_R = 3.3$  and  $f_T = -5$  to estimate the current rate of change,  $dC/dt$ .